

MATHEMATICS PAPER 20: ANSWER KEY

Unit 6 sections 6.1 and 6.2 (27 marks) + Revision: Unit 11 probability (11 marks) · 38 marks

SECTION A (10 × 1 = 10)

Q	Ans	Why
1	C	π radians = 180°
2	A	$l = r\theta$, book 6.2.1
3	B	$180^\circ = \pi$ rad, so $1^\circ = \pi/180$
4	D	Both formulas require θ in radians
5	C	$45 \times \pi/180 = \pi/4$
6	A	That is the definition of one radian
7	B	$60 + 30/60 = 60.5^\circ$
8	C	One head out of two equally likely outcomes, so $1/2$
9	D	{2,4,6} out of 6, so $3/6 = 1/2$
10	A	$P(\bar{E}) = 1 - 0.3 = 0.7$

Answer letters: **C A B D C A B C D A** (A = 3, B = 2, C = 3, D = 2, none consecutive)

SECTION B (3 × 4 = 12)

2. (i) $150 \times \pi/180 = 5\pi/6$ radians. (1 for the method, 1 for the answer)

(ii) $(7/40) \times 180 = 1260/40 = 31.5^\circ = 31^\circ 30' 0''$. (1 for reaching 31.5° , 1 for writing it as $D^\circ M'S''$)

3. (i) $l = r\theta = 10 \times 2\pi/5 = 4\pi \approx 12.57$ cm. (1 formula + substitution, 1 answer. Half off if the unit is missing.)

(ii) $A = \frac{1}{2}r^2\theta = \frac{1}{2} \times 100 \times 2\pi/5 = 20\pi \approx 62.83$ cm². (1 formula + substitution, 1 answer. Half off if the unit is missing or written cm instead of cm².)

4. (i) One radian is the angle subtended at the centre of a circle by an arc whose length is equal to the radius of that circle. **(1)**

(ii) $l = r\theta$ and $A = \frac{1}{2}r^2\theta$, where l is arc length, A is the area of the sector, r is the radius and θ is the central angle, which must be in radians. (1 each formula, 1 for the letters and the radian condition.)

SECTION C (2 × 8 = 16)

5. (i) $270 \times 3.1416/180 = 1.5 \times 3.1416 = 4.7124$ radians. **(2)**

(ii) $75^\circ 30' 45'' = 75 + 30/60 + 45/3600 = 75.5125^\circ$; $75.5125 \times 3.1416/180 = 1.3179$ radians (4 d.p.). **(2)** (1 for 75.5125° , 1 for the radian value. Accept 1.3179 or 1.3180.)

(iii) $\theta = l/r = 22/14 = 11/7 \approx 1.5714$ radians.

$A = \frac{1}{2}r^2\theta = \frac{1}{2} \times 14^2 \times 11/7 = \frac{1}{2} \times 196 \times 11/7 = 98 \times 11/7 = 154$ cm².

(1 rearranging to $\theta = l/r$, 1 the angle, 1 substituting into the area formula, 1 the answer with cm². The area is exactly 154, no π is needed.)

6. (i) Probability is a measure of how likely an event is to happen, found as the number of favourable outcomes divided by the total number of possible outcomes. It is never less than 0 and never more than 1. **(2)**

(ii) Total balls = $5 + 4 + 3 = 12$. (a) $P(\text{red}) = 5/12$ (b) $P(\text{not red}) = 7/12$ (c) $P(\text{blue}) = 3/12 = 1/4$. **(3, one each)**

(iii) (a) greater than 4 means {5, 6}, so $2/6 = 1/3$. (b) prime means {2, 3, 5}, so $3/6 = 1/2$. **(2, one each)**

(iv) $P(\bar{E}) = 1 - 2/5 = 3/5$. **(1)**

Marking notes

Unit missing on any length or area answer: half a mark, applied the same way every time. Q.5(iii) is the one to watch: she must rearrange $l = r\theta$ into $\theta = l/r$ first. If she puts 22 straight into the area formula as θ , the method mark goes but the answer mark does not. Q.6 is the re-test. On Maths exam-19 the probability question was

not attempted at all

, the single largest loss on that paper. Any genuine attempt here, even partly wrong, is progress worth saying out loud.